

1/8/26.

Class Test - 01

- 1) Reinforcement learning (RL) is a type of machine learning in which an agent learns by interacting with an environment and receiving rewards or penalties.

Main objective:-

To learn an optimal policy that maximizes the cumulative reward over time.

- 2) A Markov Decision Process (MDP) is a mathematical model used to represent decision-making problems where outcomes depend on the current state and action.

Components:-

- States (S)
- Actions (A)
- Transition Probability (P)
- Reward function (R)
- Discount factor (γ).

- 3) Bellman equation for the state value function.

$$V(S) = \sum_a \pi(a|S) \sum_{S'} P(S'|S, a) [r + \gamma V(S')]$$

where, $V(S)$ = State Value.

γ = discount factor.

4) Exploration	Exploitation.
→ Tries new actions to gain knowledge. → Helps discover better rewards.	→ Chooses the best known action. → Maximizes immediate reward using current knowledge.

5) The Multi-Armed Bandit Problem is a decision making problem where an agent chooses among several actions (slot machines) to maximize total reward while balancing exploration and exploitation.

6) Dynamic programming (DP) is a method for solving complex problems by breaking them into smaller subproblems.

Applications in RL:-

- Policy Evaluation.
- Policy Improvement
- Policy Iteration.
- Value Iteration.

7) Temporal Difference (TD) learning is a method that learns value estimates using the difference b/w consecutive predictions.

TD(0) update equation:-

$$V(S) \leftarrow V(S) + \alpha (r + V(S') - V(S))$$

here!

α = learning rate, r = Reward, γ = Discount factor.

8)	SARSA	Q-learning
	→ on-policy algorithm.	→ off-policy algorithm.
	→ learns from the action actually taken.	→ learns from the best possible action.
	→ more conservative & safer.	→ faster but may take more risks.

9) A Deep Q-network (DQN) is a reinforcement algorithm that uses a deep neural network to approximate Q-values instead of storing them in a Q-table.

Advantages over Tabular Q-learning.

- Handles large or continuous State Spaces.
- Generalizes to unseen states.

10) A Double Deep Q-network (DDQN) is an improved version of DQN that uses two networks to reduce overestimation of Q-values.

Why it is preferred over DQN:-

- Reduces overestimation bias.
- Provides more accurate Q-value estimates.